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Hyperconvexity and domains of discontinuity for surface groups in $SL_3\mathbb{C}$

BY

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THESIS [PhD candidates may use DISSERTATION here instead]

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Accessibility Statement

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Dedication

For my grandmother Martha, who also saw the joy in geometry.

Acknowledgements

Emily (integrity, knowledge, high standards)
Sam (love, support, meals, no pressure)
Family (you don't know why I wanted to do this, but that's ok)
Friends (kept me company, kept me happy)
Other math teachers and mentors
Gertrude (tiniest companion)

Preface

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List of Abbreviations

- $\mathbb{R}$ The real numbers
- $\mathbb{C}$ The complex numbers

Summary

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CHAPTER 1

Introduction

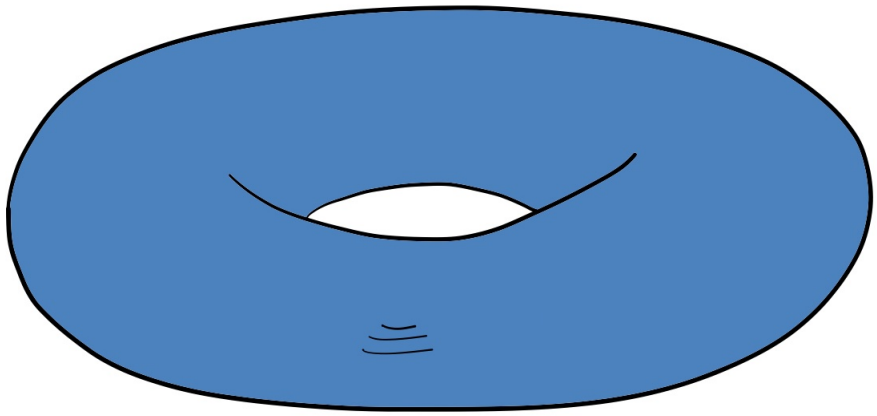

FIGURE 1. This is a torus

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & A' \arrow[r, "f'"] \arrow[d, "a"'] & B' \arrow[r, "g'"] \arrow[d, "b"'] & C' \arrow[r] \arrow[d, "c"'] & 0 \\
0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "a''"'] & B \arrow[r, "g"] \arrow[d, "b''"'] & C \arrow[r] \arrow[d, "c''"'] & 0 \\
0 \arrow[r] & A'' \arrow[r, "f''"] \arrow[d] & B'' \arrow[r, "g''"] \arrow[d] & C'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

FIGURE 2. The Snake Lemma

This is the introduction chapter. We cite some classic works [**3, 4**].

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Sed porttitor urna ac ipsum rhoncus, vel dictum risus commodo. Integer ultrices dui sit amet pulvinar tempor. Phasellus nec tellus eu turpis sodales accumsan. Proin ultr

THEOREM 1.1. *This is a theorem*

We reference Theorem 1.1.

University of Illinois Chicago.

1. Motivation

1.1. Historical context. A brief overview of how the problem developed.

$$\int_0^1 f(x)dx = 2 \tag{1}$$

How to solve (1)

1.2. Open questions. Some questions remain open for future work.
Note: I have not tested the accessibility of this table.

Monkeys	Lions
100	200

Table 1. Example Table

CHAPTER 2

Background

This chapter gives necessary background.

1. Group theory

A group is a set G with a binary operation satisfying closure, associativity, identity, and inverses.

THEOREM 2.1. *Every finite subgroup of the multiplicative group of a field is cyclic.*

PROOF. This is a standard result from algebra. $\square$

CHAPTER 3

Main Result

1. Richardson varieties in $\mathcal{F}^3$

TODO: This maybe belongs in an introduction / background section instead.

For any flag $x \in \mathcal{F}^3$, we define $Sch(x)$ the principle Schubert varieties associated with x. Let $x = (l, H)$ where $l \in \mathbb{CP}^2$ and $H \in (\mathbb{CP}^2)^*$ such that $l_x \subset H_x$. We distinguish the elements of $Sch(x)$ using elements of the Weyl group $W(SL_3\mathbb{C})$ as follows:

- $\alpha_{Id}(x) = \{(l, H)\} = \{x\}$,
- $\alpha_1(x) = \{(l, H_l) : l \subset H_l\}$,
- $\alpha_2(x) = \{(l_H, H) : l_H \subset H\}$,
- $\alpha_{12}(x) = \{(k, H_l) : l \subset H_l\}$,
- $\alpha_{21}(x) = \{(l_H, G) : l_H \subset H\}$, and
- $\alpha_{121}(x) = \alpha_{212}(x) = \mathcal{F}^3$.

The principle Schubert varieties are closed, and have complex dimensions $(0, 1, 1, 2, 2, 3)$, respectively.

We call a pair of flags $x, y \in \mathcal{F}^3$ *transverse* if they are in general position, meaning that for any proper, principle Schubert variety α, then $x \notin \alpha(y)$ and $y \notin \alpha(x)$.

Let $x, y \in \mathcal{F}^3$ transverse. Define $R^0_{x,y}$ to be their irreducible 0-dimensional Richardson varieties, which are the intersections between principle Schubert varieties $\alpha(x) \cap \beta(y)$ where $dim(\alpha)+dim(\beta) = dim(\mathcal{F}^3)$. There are exactly six such varieties $\{x, x_y, x^y, y_x, y^x, y\}$, where:

- $x := \{x\} \cap \alpha_{121}(y)$,
- $x_y := \alpha_1(x) \cap \alpha_{12}(y)$,
- $x^y := \alpha_2(x) \cap \alpha_{21}(y)$,
- $y_x := \alpha_{12}(x) \cap \alpha_1(y)$,
- $y^x := \alpha_{21}(x) \cap \alpha_2(y)$, and
- $y := \alpha_{121}(x) \cap \{y\}$.

Given two tranverse flags, the Richardson 0-varieties can be computed explicitly:

Example 3.1. *Choose x and y are the attracting and repelling flags of $\rho(g) = diag(\lambda, \mu, \nu)$ where $|\lambda| > |\mu| > |\nu|$. Then the six Richardson 0-varieties in $R^0_{x,y}$ are $x = (e_1, e_1 + e_2)$, $x_y = (e_1, e_1+e_3)$, $x^y = (e_2, e_1+e_2)$, $y_x = (e_3, e_1+e_3)$, $y^x = (e_2, e_2+e_3)$, and $y = (e_3, e_2+e_3)$.*

Let $x, y \in \mathcal{F}^3$ transverse, and define $R^1_{x,y}$ to be irreducible 1-dimensionsal Richardson varieties. Similarly, these are intersections of principle Schubert varieties $\alpha(x) \cap \beta(y)$ where $dim(\alpha)+dim(\beta) = dim(\mathcal{F}^3)+1$. There are six such 1-varieties, each homeomorphic to $\mathbb{CP}^1$, which we distinguish using pairs of elements of $R^0_{x,y}$.

- $span(x, x_y) := \alpha_1(x) \cap \alpha_{121}(y)$,
- $span(x, x^y) := \alpha_2(x) \cap \alpha_{121}(y)$,
- $span(x_y, y_x) := \alpha_{12}(x) \cap \alpha_{12}(y)$,
- $span(x^y, y^x) := \alpha_{21}(x) \cap \alpha_{21}(y)$,
- $span(y_x, y) := \alpha_{121}(x) \cap \alpha_1(y)$, and
- $span(y^x, y) := \alpha_{121}(x) \cap \alpha_2(y)$.

The 1-varieties are not pairwise disjoint, e.g. $\{x\} = span(x, x_y) \cap span(x, x^y)$. The points of intersection between 1-dimensional Richardson varieties are the 0-dimensional Richardson varieties.

Remark 3.1. *If $x, y \in \mathcal{F}^3$ transverse and α, β principle Schubert varieties such that $dim(\alpha) + dim(\beta) < dim(\mathcal{F}^3)$, then $\alpha(x) \cap \beta(y) =$.*

2. Assumptions and notation

Let Γ the fundamental group of a closed hyperbolic surface. Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ Anosov. We use the following notation related to ρ and ρ'.

- $\xi, \xi' : \partial\Gamma \to \mathcal{F}^3$ refers to the continuous, transverse, ρ- or ρ'- equivariant, dynamics-preserving boundary maps.
- $\Lambda, \Lambda' \subset \mathcal{F}^3$ refers to the limit curves, which are the images of ξ, ξ'.
- $Th(x) = \alpha_1(x) \cup \alpha_2(x)$ refers to the balanced ideal thickening of $x \in \mathcal{F}^3$.
- $Th(\Lambda) = \cup_{x\in\Lambda} Th(x)$ refers to the thickening of the limit curve.

We call ρ, ρ' *topologically conjugate* if there exists a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that for all $g \in \Gamma$ and all $x \in \mathcal{F}^3$, then $F(\rho(g) \cdot x) = \rho'(g) \cdot F(x)$. We call F the *topological conjugacy* between ρ and ρ'.

Recall that ρ is *hyperconvex* if for all $x = (l_x, H_x), y = (l_y, H_y), z = (l_z, H_z) \in \Lambda$, then $l_z \not\subset l_x + l_y$ and $(H_x \cap H_y) \not\subset H_z$.

3. Hyperconvexity is closed under topological conjugacy

THEOREM 3.1. *Hyperconvexity is closed under topological conjugacy. In other words, if ρ and ρ' are topologically conjugate, then ρ is hyperconvex if and only if ρ' is hyperconvex.*

We use two supporting lemmas to prove this theorem. First, we show that the relative position of 0-dimensional Richardson varieties of Λ determines whether ρ is hyperconvex.

LEMMA 3.1.1. *An Anosov representation ρ is hyperconvex* if and only if *for all $x, y, z \in \Lambda^{(3)}$ we have $R^0_{x,y} \cap R^0_{x,z} = \{x\}$.*

For the second lemma, assume that $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'. Clearly $F(\Lambda) = \Lambda'$. This mapping is pointwise, i.e. for all $\alpha \in \partial\Gamma$, then $F(\xi(\alpha)) = \xi'(\alpha)$. So for $x \in \Lambda$, we can use the notation $x' := F(x) \in \Lambda'$.

Topological conjugacies may preserve the 0- and 1-dimensional Richardson varieties of Λ in one of two ways. We say F *matches* Richardson varieties if for all $x, y \in \Lambda^{(2)}$, then $F(x_y) = x'_{y'}$ and $F(span(x, x_y)) = span(x, x'_{y'})$. We say F *switches* Richardson varieties if for all $x, y \in \Lambda^{(2)}$, then $F(x_y) = x'^{y'}$ and $F(span(x, x_y)) = span(x, x'^{y'})$.

LEMMA 3.1.2. *For all $x, y \in \Lambda^{(2)}$, then $F(R^0_{x,y}) = R^0_{x',y'}$ and $F(R^1_{x,y}) = R^1_{x',y'}$. Furthermore, F either matches or switches Richardson varieties.*

We also have the following consequence of Lemma 3.1.2, which is of independent interest:

COROLLARY 3.1.1. *For all $x \in \Lambda$, then $F(Th(x)) = Th(x')$.*

4. Proof of Theorem 3.1.

We first prove our main theorem using our supporting lemmas.

PROOF. (Theorem 3.1) Suppose $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between Anosov representations $\rho, \rho' :\to SL_3\mathbb{C}$.

We begin with the forward direction. Assume that ρ is hyperconvex. Let $x, y, z \in \Lambda^{(3)}$. By Lemma 3.1.1, then $x_y \neq x_z$ and $x^y \neq x^z$. By Lemma 3.1.2, then the sets

$$\{F(x_y), F(x^y), F(x_z), F(x^z)\} = \{x'_{y'}, x'^{y'}, x'_{z'}, x'^{z'}\} \tag{2}$$

contain four distinct points, and in particular $x'_{y'} \neq x'_{z'}$ and $x'^{y'} \neq x'^{z'}$. Since x, y, z were general, then this holds for all triples in $x', y', z' \in \Lambda'$, and ρ' is hyperconvex.

The other direction proceeds similarly. Assume that ρ is *not* hyperconvex. By Lemma 3.1.1, there exist $x, y, z \in \Lambda^{(3)}$ such that $x_y = x_z$ or $x^y = x^z$. By Lemma 3.1.2, then F either matches or switches Richardson varieties. So then $x'_{y'} = x'_{z'}$ or $x'^{y'} = x'^{z'}$. Therefore ρ' is *not* hyperconvex. □

We also prove our lemma linking Richardson 0-dimensional varieties to hyperconvexity.

Proof. (Lemma 3.1.1) Suppose ρ is Anosov. Let $x = (l_x, H_x), y = (l_y, H_y), z = (l_z, H_z) \in \Lambda^{(3)}$. Since x, y, z are pairwise transverse, then

$$y_x, y^x, z_x, z^x \notin R^0_{x,y} \cap R^0_{x,z}. \tag{3}$$

It suffices to check the pairs $x_y, x_z \in \alpha_1(x)$ and $x^y, x^z \in \alpha_2(x)$.

By definition, $x_y = (l_x, l_x + l_y)$ and $x_z = (l_x, l_x + l_z)$, which are distinct if and only if $l_x + l_y \neq l_x + l_z$, or equivalently if and only if $l_z \not\subset l_x + l_y$.

Similarly, $x^y = (H_x \cap H_y, H_x)$ and $x^z = (H_x \cap H_z, H_x)$, which are distinct if and only if $H_x \cap H_y \neq H_x \cap H_z$, or equivalently if and only if $H_x \cap H_y \not\subset H_z$.

So $R^0_{x,y} \cap R^0_{x,z} = \{x\}$ for all $x, y, z \in \Lambda^{(3)}$ is equivalent to ρ hyperconvex. □

5. Action of $\rho(g)$ on F^3

The remainder of this chapter is devoted to the proof of Lemma 3.1.2.

In this section we describe the action of $\rho(g)$ on $\mathcal{F}^3$ for $g \in \Gamma$ infinite order. Since ρ is Anosov, then $\rho(g)$ is strongly loxodromic, i.e. its eigenvalues have distinct magnitude. That means $\rho(g)$ has attracting and repelling limit curve points $x = \xi(g^+)$ and $y = \xi(g^-)$, respectively.

Under the action $\rho(g)$, most of $\mathcal{F}^3$ is attracted to x and repelled by y. We define the *uphill* of x as the subset attracted to x under $\rho(g)$:

$$U_{\rho(g)}(x) := \{f \in F^3 : \rho(g^n) \cdot f \to x\}. \tag{4}$$

Similarly, we define the *downhill* of y as the subset repelled by y under $\rho(g)$, or equivalently the subset attracted by y under $\rho(g^{-1})$:

$$D_{\rho(g)}(y) = U_{rho(g^{-1})}(y) := \{f \in \mathcal{F}^3 : \rho(g^{-n}) \cdot f \to y\}. \tag{5}$$

We then define the *exceptional set* of $\rho(g)$ as the subset *not* attracted to x or *not* repelled by y under $\rho(g)$:

$$E_{\rho(g)} = \mathcal{F}^3 \setminus (U_{\rho(g)}(x) \cap D_{\rho(g)}(y)). \tag{6}$$

The set $E_{\rho(g)}$ is nonempty and closed, since it is complement of an open set. It contains the fixed points of $\rho(g) \cdot \mathcal{F}^3$, which are precisely $R^0_{x,y}$. It suffices to show this for $x, y, \rho(g)$ as in Example 3.1, since up to conjugation $\rho(g) = diag(\lambda, \mu, \nu)$.

We can partition $E_{\rho(g)}$ using each element's attracting and repelling fixed points. By taking the closure of these subsets, we recover the 1-dimensional Richardson varieties $R^1_{x,y}$.

- $span(x, x_y) = \alpha_1(x) = \overline{U(x) \cap D(x_y)}$
- $span(x, x^y) = \alpha_2(x) = \overline{U(x) \cap D(x^y)}$
- $span(x_y, y_x) = \overline{U(x_y) \cap D(y_x)}$
- $span(x^y, y^x) = \overline{U(x^y) \cap D(y^x)}$

- $span(y_x, y) = \alpha_1(y) = \overline{U(y_x) \cap D(y)}$
- $span(y^x, y) = \alpha_2(y) = \overline{U(y^x) \cap D(y)}$

So $E_{\rho(g)}$ is homeomorphic to the union of six copies of $\mathbb{CP}^1$, each identified with two neighbors in a cyclic order $(x, x_y, y_x, y, y^x, x^y)$.

6. F preserves Richardson 0- and 1-varieties of $\xi(g^+), \xi(g^-)$

Assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$. We use the notation $x' = F(x), y' = F(y)$.

We first consider the attracting and repelling flags for an infinite order element $g \in \Gamma$.

Proposition 3.1. *Let $x, y \in \Lambda$ such that $x = \xi(g^+)$ and $y = \xi(g^-)$. Then all of the following are true:*

(1) $F(R^0_{x,y}) = R^0_{x',y'}$,
(2) $F(R^1_{x,y}) = R^1_{x',y'}$, *and*
(3) F *either matches or switches the Richardson varieties of x and y.*

Proof. Since F is a topological conjugacy, it must preserve the exceptional set of $\rho(g)$, so $F(E_{\rho(g)}) = E_{\rho'(g)}$. Similarly, since F preserves the limit curve pointwise, then $F(x) = x'$ and $F(y) = y'$.

Since $F|_{E_{\rho(g)}}$ is a homeomorphism, it sends 1-varieties to 1-varieties, and sends intersections of elements of $R^1_{x,y}$ to intersections of elements of $R^1_{x',y'}$. This proves (1) and (2).

To prove (3), we observe that:

$$F(x) = F(span(x, x_y) \cap span(x, x^y)) \tag{7}$$
$$= F(span(x, x_y)) \cap F(span(x, x^y)) \tag{8}$$
$$= span(x', x'_{y'}) \cap span(x', x'^{y'}) \tag{9}$$
$$= x'. \tag{10}$$

So we conclude that either $F(x_y) = x'_{y'}$ and $F(span(x, x_y)) = span(x', x'_{y'})$, or $F(x_y) = x'^{y'}$ and $F(span(x, x_y)) = span(x', x'^{y'})$. □

7. F preserves Richardson 0-varieties of Λ

In this section, we still assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ, ρ'. We extend our argument from the Richardson 0-varieties of pairs $\xi(g^+), \xi(g^-)$ to the Richardson 0-varieties of all distinct pairs in Λ.

First, we extend Proposition 3.1 to Richardson varieties of pairs x, z where x is the attracting fixed point of an infinite order element $g \in \Gamma$ and z is a (distinct) generic point on Λ.

Proposition 3.2. *Let $g \in \Gamma$ infinite order, and define $x := \xi(g^+)$. Let $z \in \Lambda \setminus \{x\}$. Then $F(x_z) \in \{x'_{z'}, x'^{z'}\}$.*

We use the following result from group theory:

REMARK 3.2 (TODO REFERENCE). *Let Γ hyperbolic with boundary $\partial\Gamma \equiv S^1$, and let $x, z \in \partial\Gamma$. Then there exists a quasigeodesic sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that for all $y \in \partial\Gamma$:*

$$lim_{i\to\infty} g_i \cdot y = \begin{cases} x : y = x \\ z : y \neq x \end{cases} . \tag{11, 12}$$

PROOF. (Proposition 3.2) If $z = \xi(g^-)$ then we are done by Prop 3.1, so assume not. Set $y := \xi(g^-)$.

Let $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ as in Rem. 3.2, such that $\lim_{i=1} \rho(g_i) \cdot y = z$ and $lim_{i=1} \rho(g_i) \cdot x = x$. Let $x_i := \rho(g_i) \cdot x$ and $y_i := \rho(g_i) \cdot y$, and let $x'_i := F(x_i), y'_i := F(y_i)$.

By Prop. 3.1 then for all $i \in \mathbb{N}$ we have $F(x_{iy_i}) \in \{x'_{iy'_i}, x'^{y'_i}_i\}$. Clearly $\lim_{i\to\infty} x_{iy_i} = x_z$, $\lim_{i\to\infty} x'_i = x'$ and $\lim_{i\to\infty} y'_i = z'$. By continuity of F and compactness of $\mathcal{F}^3$, then

$$\begin{aligned} F(x_z) &= F(\lim_{i\to\infty} x_{iy_i}) && (13)\\ &= \lim_{i\to\infty} F(x_{iy_i}) && (14)\\ &\in \{\lim_{i\to\infty} x'_{iy'_i}, \lim_{i\to\infty} x'^{y'_i}_i\} && (15)\\ &= \{x'_{z'}, x'^{z'}\}. && (16) \end{aligned}$$

□

Next, we extend Proposition 3.2 to Richardson varieties of the form w_z where both w and z are generic points of Λ.

PROPOSITION 3.3. *For all $w, z \in \Lambda^{(2)}$, then $F(w_z) \in \{w'_{z'}, w'^{z'}\}$.*

PROOF. If w is the attracting fixed point of an infinite order element, we are done by Prop 3.2, so assume not.

Let $\{x_i\}_{i=1} \in \Lambda^{\mathbb{N}}$ such that each $x_i = \xi(g_i)$ the attracting fixed point of an infinite order word, and $\lim_{i\to\infty} x_i = w$. This sequence exists because attracting fixed points of $\rho(g)$ (for $g \in \Gamma$ infinite order) are dense in Λ. Define $x'_i = F(x_i)$. By continuity of F, we have

$$\begin{aligned} F(w) &= F(\lim_{i\to\infty} x_i) && (17)\\ &= \lim_{i\to\infty} F(x_i) && (18)\\ &= \lim_{i\to\infty} (x'_i) && (19)\\ &= w'. && (20) \end{aligned}$$

Then by Prop 3.2 and by continuity of F, we have

$$\begin{aligned} F(w_z) &= F(\lim_{i\to\infty} x_{iz}) && (21)\\ &= \lim_{i\to\infty} F(x_{iz}) && (22)\\ &\in \{\lim_{i\to\infty} (x'_{iz'}), \lim_{i\to\infty} (x'^{z'}_i)\} && (23)\\ &= \{w'_{z'}, w'^{z'}\}. && (24) \end{aligned}$$

□

8. F preserves Richardson 1-varieties of Λ

In this section, we highlight how F preserves the Richardson 1-varieties of Λ, in particular the balanced ideal thickening centered at each point $x \in \Lambda$.

Observe that $Th(x) = \alpha_1(x) \cup \alpha_2(x) = span(x, x_y) \cup span(x, x^y)$ for some $y \in \Lambda \setminus \{x\}$.

Proof. (Corollary 3.1.1) By Prop. 3.1, $\{F(x_y), F(x), F(x^y)\} = \{x'_{y'}, x, x'^{y'}\}$. Also, Observe that F is a homeomorphism that preserves dynamics in that $F(U_{\rho(g)}) = U_{\rho'(g)}$ and $F(D_{\rho(g)}) = D_{\rho'(g)}$. So then

$$F(Th(x)) = F(span(x, x_y) \cup span(x, x^y)) \tag{25}$$
$$= span(x', x'_{y'}) \cup span(x', x'^{y'}) \tag{26}$$
$$= Th(x'). \tag{27}$$

□

If F *matches* Richardson varieties, then for all $x \in \Lambda$ we have $F(x_y) = x'_{y'}$ and $F(\alpha_1(x)) = \alpha_1(x')$. Otherwise, if F *switches* Richardson varieties, then for all $x \in \Lambda$ we have $F(x_y) = x'^{y'}$ and $F(\alpha_1(x)) = \alpha_2(x')$.

Corollary 3.1.2. *F either matches or switches Richardson varieties.*

Proof. Suppose for some $w \in \Lambda$ we have $F(\alpha_1(w)) = \alpha_1(w')$. Let $z \in \Lambda\ \{w\}$. Since F is a homeomorphism, by Cor. 3.1.1 we have $F(\alpha_1(z)) \in \{\alpha_1(z'), \alpha_2(z')\}$.

For contradiction, suppose that $F(\alpha_1(z)) = \alpha_2(z')$. Then for all $g \in \Gamma$, by the definition of a topological conjugacy, we have $F(\alpha_1(\rho(g) \cdot z)) = \alpha_2(\rho'(g) \cdot z')$.

Since $\rho(g) \cdot z$ is dense in Λ, then there exists a sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that $\lim_{i\to\infty} \rho(g_i) \cdot z = w$. Define $z_i = \rho(g_i) \cdot z$.

Then

$$\alpha_1(w') = F(\alpha_1(w)) \tag{28}$$
$$= F(\alpha_1(\lim_{i\to\infty} z_i)) \tag{29}$$
$$= \lim_{i\to\infty} F(\alpha_1(z_i)) \tag{30}$$
$$= \lim_{i\to\infty} \alpha_2(w) \tag{31}$$
$$= \alpha_2(w). \tag{32}$$

This is a contradiction. □

Clearly $x_y \in \alpha_1(x)$, so if F matches Richardson varieties then $F(x_y) = x'_{y'}$, and if F switches Richardson varities then $F(x_y) = x'^{y'}$.

So for all $x, y \in \Lambda$, the topological conjugacy F sends $R^0_{x,y}$ to $R^0_{x',y'}$, sends $R^1_{x,y}$ to $R^1 x', y'$, and either matches or switches Richardson varieties. This completes the proof of Lemma 3.1.2.

9. Additional remarks

Remark 3.3. *For distinct $x, y \in \mathcal{F}^3 \setminus \Lambda$, F will not necessarily preserve the 0-dimensional Richardson varieties.*

Remark 3.4. *Theorem 3.1 applies to virtual hyperbolic surface groups as well, e.g.* $\Gamma = \pi_1(\mathcal{O}^2)$ *where* $\mathcal{O}^2$ *is a hyperbolic 2-orbifold.*

Conjectural picture for higher dimensions:

I suspect that topological conjugacies of Borel Anosov representations will preserve Richardson varieties up to $R^{n-2}_{x,y}$ organized into a cell decomposition with data an $(n-1)$-valent graph where vertices correspond to 0-varieties, edges correspond to 1-varieties, fully connected triples correspond to 2-varieties, etc. Matching/switching Richardson varieties will mean sending Richardson varieties to their dual.

I think balanced ideals I correspond to subgraphs of the Richardson variety graph containing exactly half of the vertices. In other words, $Th^I(x) \cap Th^I(y) =$, and

$$(Th^I(x) \cap R^0_{x,y}) \cup (Th^I(y) \cap R^0_{x,y}) = R^0_{x,y}. \tag{33}$$

However, topological conjugacies should not depend on the choice of balanced ideal, so then they determine diffeomorphisms between all types of quotient manifolds. In other words, if $F : \mathcal{F}^n \to \mathcal{F}^n$ a topological conjugacy between ρ, ρ', then:

$$F(M^I_\rho) = M^I_{\rho'} \text{ for all balanced ideals } I \subset W(SL_n\mathbb{C}). \tag{34}$$

Most importantly, I believe the conclusions of Theorem 3.1 will apply in higher rank:

Conjecture 3.1. *For* $\rho, \rho' : \Gamma \to SL_n\mathbb{C}$ *Borel Anosov and topologically conjugate. Then* ρ *hyperconvex if and only if* ρ' *hyperconvex.*

CHAPTER 4

Another result

1. Notation

2. Anosov components have one topological conjugacy class

THEOREM 4.1. *Suppose $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ in the same Anosov component. Suppose furthermore that ρ and ρ' are irreducible. Then ρ, ρ' are topologically conjugate.*

Combined with Theorem 3.1, we get the following corollaries:

COROLLARY 4.1.1. *All quasi-Hitchin representations are hyperconvex. All quasi-Barbot representations are* not *hyperconvex.*

Furthermore:

COROLLARY 4.1.2. *The quasi-Hitchin and quasi-Barbot Anosov components are disjoint.*

Additionally, distinct Anosov components may have the same topological conjugacy class. For example:

PROPOSITION 4.1. *Let $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov and let ρ^* its dual. Then ρ, ρ^* are topologically conjugate.*

We also consider the topological conjugacies of reducible representations $\mathcal{A}_{red}$.

THEOREM 4.2. *There are exactly three topological conjugacy classes of $\mathcal{A}_{red}(\Gamma)$:*

- $[\rho_{ss}]$
- $[\rho_{line}] = [\rho_{plane}]$
- $[\rho_{dense}]$

3. Key ingredients for proof

In this section, we highlight prior results that are used in the proof of Theorem 4.1. First, we use the following theorem of Stecker:

THEOREM 4.3 (**??**, Thm 3.24). *Let $\rho : \Gamma \to G$ Borel Anosov. If $dim(\Lambda) \leq mbic(G) - 2$, then every non-empty cocompact domain of discontinuity in $\mathcal{F}^3$ is dense and connected, and there is a bijection:*

$$\{\textit{balanced ideals in } W(G)\} \leftrightarrow \{\textit{nonempty cocompact domains of discontinuity in } \mathcal{F}(G)\} \tag{35}$$

We tailor Stecker's theorem to our assumptions. Recall that $G = SL_3\mathbb{C}$, so $mbic(G) = 4$ and $\mathcal{F}(G) = \mathcal{F}^3$. Then $W(G) = A_2$ which has exactly one balanced ideal. In $SL_3\mathbb{C}$, all Anosov representations are Borel Anosov. Furthermore, we assume that Γ is a hyperbolic surface group, so $\partial\Gamma \cong S^1$. Then $1 \leq dim(\Lambda_\rho) \leq 2$, because ξ_ρ is an injective and continuous function of S^1. So the condition in Theorem 4.3 is met, and:

REMARK 4.1. *For Γ a hyperbolic surface group and $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov, then ρ has a* unique *nonempty cocompact domain of discontinuity $\Omega_\rho \subset \mathcal{F}^3$. Furthermore, $\Omega_r ho$ is connected and dense in $\mathcal{F}^3$.*

We also use the following lemma, proved in the following section:

LEMMA 4.3.1. *For Γ a hyperbolic surface group and $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov, then Ω_ρ is simply connected.*

Finally, we use the following theorem of Dumas and Sanders:

THEOREM 4.4 (**??**, Thm 5.1). *Let (P, Q) be parabolic subgroups of G, with P symmetric, and let $I \subset W(G)$ be a balanced ideal of type (P, Q). Let $\rho : \Gamma \to G$ be a P-Anosov representation. Then for any $\rho' \in \mathcal{A}(\rho, P)$, the quotient manifolds M_ρ^I and $M_{\rho'}^I$ are diffeomorphic.*

We again tailor this theorem to our assumptions that $G = SL_3\mathbb{C}$, where there is only symmetric parabolic subgroup and one balanced ideal in $W(G)$. Then:

REMARK 4.2. *For Γ a hyperbolic surface group and $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ in the same Anosov component, then*

$$M_\rho \equiv_{diffeo} M_{\rho'}. \tag{36}$$

4. Proof of Theorem 4.1

PROOF. (Theorem 4.1) Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ irreducible and in the same Anosov component. We want to show the existence of $F : \mathcal{F}^3 \to \mathcal{F}^3$ a topological connjugacy of ρ, ρ'.

Then by Thm. 4.4, there exists a diffeomorphism $D : M_\rho \to M_{\rho'}$. This lifts to a diffeomorphism of the universal cover $\tilde{D} : \tilde{M}_\rho \to \tilde{M}_{\rho'}$. Then by Lemma 4.3.1, the domains of discontinuity Ω_ρ, $\Omega_{\rho'}$ are simply connected, so $\tilde{M}_\rho \equiv \Omega_\rho$ and $\tilde{M}_{\rho'} \equiv \Omega_{\rho'}$. So we can consider $\tilde{D} : \Omega_\rho \to \Omega_{\rho'}$.

Clearly $\tilde{D}$ is a topological conjugacy of Ω_ρ and $\Omega_{\rho'}$, since it is a diffeomorphism and for all $f \in \Omega_\rho$ and $g \in \Gamma$ we have $\tilde{D}(\rho(g) \cdot f) = \rho'(g) \cdot \tilde{D}(f)$.

Next, we want to extend $\tilde{D}$ to a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that $F|_{\Omega_\rho} \equiv \tilde{D}$. By Theorem 4.3, Ω_ρ is an open, connected dense subset of $\mathcal{F}^3$, so $\tilde{D}$ has a unique continuous extension to $\mathcal{F}^3$ using sequences in Ω_ρ which limit to points in $Th(\Gamma)$.

TODO check thet F is a topological conjugacy. TODO where do we use irreducible? □

REMARK 4.3. *The topological conjugacy F may not be unique, however it is uniquely determined by the choice of the diffeomorphism D and its lift $\tilde{D}$.*

PROOF. (Lemma 4.3.1) Fact: $\pi_1\mathcal{F}^3$ is trivial. Fact: The odd homology groups $H^{2n+1}(\Omega_\rho)$ vanishes. Fact: Then $\pi_1(\Omega_\rho)$ is perfect, since $\pi_1(\Omega_\rho)^{ab} = H^1(\Omega_\rho) = \{e\}$.

Claim: First consider $\Omega_{\rho_{Fuchs}}$ where ρ_F is $SL_3\mathbb{C}$-Fuchsian. Then $\Omega_{\rho_{Fuchs}}$ is simply connnected.

Claim: If there is a transverse homotopy between $\Lambda_{\rho_{Fuchs}}$ and Λ_ρ, then $\pi_1(\Omega_{\rho_{Fuchs}}) = \pi_1(\Omega_\rho)$.

Claim: There is a transverse homotopy between $\Lambda_{\rho_{Fuchs}}$ and Λ_ρ. □

5. Topological conjugacies of reducible representations

This argument for Theorem 4.2 hinges on whether ρ fixes a line and/or a plane in $\mathbb{C}^3$. Let $\rho : \Gamma \to SL_3\mathbb{C}$ reducible. We say that ρ is

(1) *semisimple* if ρ fixes a line $l \in CP^2$ and a plane $H \in (CP^2)^*$,
(2) *line-reducible* if ρ fixes a line $l \in CP^2$, but no plane,
(3) *plane-reducible* if ρ fixes a plane $H \in (CP^2)^*$, but no line, and
(4) *neither line- nor plane-reducible* otherwise.

Given $l \in \mathbb{CP}^2$, we use the abuse of notation

$$\alpha_1(l) := \alpha_1(l, H) \tag{37}$$

for any $H \supset l$ because $\alpha_1(l, H)$ does not depend on choice of H. Similarly, for $H \in (\mathbb{CP}^2)^*$, we use the analogous abuse of notation

$$\alpha_2(H) := \alpha_2(l, H) \text{ for some } l \subset H. \tag{38}$$

If ρ is semisimple, then for all $x, y, z \in \Lambda_\rho^{(3)}$ we have $x_y = x_z$, $y_x = y_z$, $z_x = z_y$, and $x^y = x^z$, $y^x = y^z$, $z^x = z^y$. So ρ fixes the line $l = l_{x^y}$ and the plane $H = H_{x_y}$. This means that the following sets are invariant under ρ:

$$\alpha_1(l) = span(x_y, y_x) = span(x_z, z_x) = span(y_z, z_y), \text{ and} \tag{39}$$

$$\alpha_2(H) = span(x^y, y^x) = span(x^z, z^x) = span(y^z, z^y). \tag{40}$$

Similarly, if ρ is line-reducible (but not semisimmple), then all $x^y = x^z$, $y^x = y^z$, $z^x = z^y$ but not all $x_y = x_z$. So ρ also fixes the line $l = l_{x^y}$, so the following set is invariant under ρ:

$$\alpha_1(l) = span(x^y, y^x) = span(x^z, z^x) = span(y_z, z_y). \tag{41}$$

If ρ is plane-reducible (but not semisimple), then for all $x, y, z \in \Lambda_\rho^{(3)}$ we have $x_y = x_z$, $y_x = y_z$, $z_y = z_x$, but not all $x^y = x^z$. So ρ fixes the plane $H = H_{x_y}$, so the following set is invariant under ρ:

$$\alpha_2(H) = span(x_y, y_x) = span(x_z, z_x) = span(y_z, z_y). \tag{42}$$

Finally, if ρ is neither line- nor plane-reducible, then ρ fixes no complex line or plane in $\mathbb{C}^3$.

Proof. (Theorem 4.2) If ρ, ρ' are in the same representation conjugacy class and have the same reducible type, then they are topologically conjugate.

Now we consider pairs $\rho, \rho' \in \overline{[\rho]}$ with different reducible types.

Semisimple representations are only topologically conjugate to other semisimple representations.

Since line-reducible and plane-reducible representations but fix a complex line in Ω_ρ, then they can be topologically conjugate.

Finally, if ρ is neither line- nor plane-reducible, then it does not fix any complex line in Ω_ρ, and is therefore not topologically conjugate to other reducible types. However, it is topologically conjugate to other irreducible representations in the same Anosov component. □

We also gain the following corollary, which is immediate from the proof of the Theorem and Lemma 3.1.2:

Corollary 4.4.1. *Suppose F a topological conjugacy between ρ line-reducible and some reducible ρ'. If ρ' line-irreducible, then F matches Richardson varieties. Otherwise ρ' is plane-irreducible and F switches Richardson varieties.*

CHAPTER 5

Vitae

For UIC this can be a full CV or a short version (e.g. half a page) with previoius degrees, work experience and publications.

APPENDIX A

Technical Lemmas

Here we collect some supporting lemmas.

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